(5) Any subspace $\mathfrak{a}$ of a Lie algebra $\mathfrak{g}$ which is closed under the commutator map $[\ ,\ ]$, i.e., such that $[a, b] \in \mathfrak{a}$ if $a, b \in \mathfrak{a}$. Such a subspace is called a **Lie subalgebra** of $\mathfrak{g}$.

Remark 2.9.3. Ado's theorem says that any finite dimensional Lie algebra is a Lie subalgebra of $\mathfrak{gl}(V)$ for a suitable finite dimensional vector space V.

Remark 2.9.4. Derivations are important because they are the "infinitesimal version" of automorphisms (i.e., isomorphisms onto itself). For example, assume that $g(t)$ is a differentiable family of automorphisms of a finite dimensional algebra A over $\mathbb{R}$ or $\mathbb{C}$ parametrized by $t \in (-\epsilon, \epsilon)$ such that $g(0) = \mathrm{Id}$. Then $D := g'(0) : A \to A$ is a derivation (check it!). Conversely, if $D : A \to A$ is a derivation, then e^{tD} is a 1-parameter family of automorphisms (give a proof!).

This provides a motivation for the notion of a Lie algebra. Namely, we see that Lie algebras arise as spaces of infinitesimal automorphisms (= derivations) of associative algebras. In fact, they similarly arise as spaces of derivations of any kind of linear algebraic structures, such as Lie algebras, Hopf algebras, etc., and for this reason play a very important role in algebra.

Here are a few more concrete examples of Lie algebras:

(1) $\mathbb{R}^3$ with $[u, v] = u \times v$, the cross-product of u and v.

(2) $\mathfrak{sl}(n)$, the set of $n \times n$ matrices with trace 0.
For example, $\mathfrak{sl}(2)$ has the basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

with relations

$$[h, e] = 2e,\ [h, f] = -2f,\ [e, f] = h.$$

(3) The Heisenberg Lie algebra $\mathcal{H}$ of matrices $\left(\begin{smallmatrix} 0 & * & * \\ 0 & 0 & * \\ 0 & 0 & 0 \end{smallmatrix}\right)$.
It has the basis

$$x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad y = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad c = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

with relations $[y, x] = c$ and $[y, c] = [x, c] = 0$.